Speed

Chapters 8 & 9 — Supplement

A units-method companion for upper-primary problem solving

Grades 4–6 | English | ThinkAlgo editorial series

What is inside

- **Chapter 8 — Fractions of a journey (the units method).** Draw a journey as equal units, find one unit, then scale up. Theory, 2 worked examples and 20 practice problems, each with a bar model.
- **Chapter 9 — Journeys with a change of speed.** There-and-back trips, partial legs, differences in average speed, and speed changes mid-journey. Theory, 2 worked examples and 20 practice problems.

This booklet is produced in a single source with two editions.

Student edition. Solutions are hidden; blank working space is provided.

Chapter 8

Fractions of a journey — the units method

Many journey problems become easy once you picture the whole trip as a row of *equal units*. If you know how far (or how long) *one* unit is, you can find the *whole* just by scaling up. That single idea powers this entire chapter.

Key idea — the units method

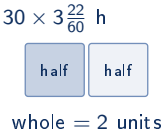
Split the journey into equal units. Choose units so the part you know is a whole number of them.

1. **Draw** the journey as equal units (a bar model).
2. **Find 1 unit** from the part you are given (for example, "7 units = 840 km, so 1 unit = 120 km").
3. **Scale up** to the whole: whole = (number of units) × (1 unit).

Two useful specials: **(a)** if you travel *half* the trip, the whole is 2× that part; **(b) average speed** $= \dfrac{\text{total distance}}{\text{total time}}$ (never just the average of the two speeds).

Example 1 — a half journey

Emma cycles for **3 h 22 min** at **30 km/h** and arrives exactly *halfway* along her route. How long is the whole route?



Solution

Write the time as hours. 3 h 22 min $= 3\frac{22}{60} = \dfrac{202}{60}$ h.

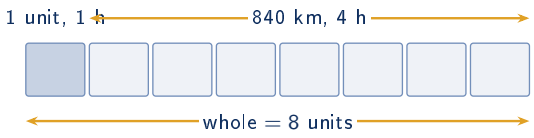
Find the half (one unit). Distance = speed × time $= 30 \times \dfrac{202}{60} = 101$ km.

Scale up to the whole. The whole route is 2 units: $2 \times 101 = 202$ km.

> **Answer:** The whole route is **202 km**.

Example 2 — units and average speed

Oliver drives $\frac{1}{8}$ **of a journey in 1 h**. The remaining **840 km** takes him **4 h**. Find the length of the whole journey and his average speed for the trip.



Solution

Count the units. The journey is 8 units; $\frac{1}{8}$ is 1 unit, the rest is 7 units.

Find 1 unit. 7 units $=$ 840 km, so 1 unit $= 840 \div 7 = 120$ km.

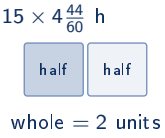
Scale up to the whole. Whole $= 8 \times 120 = 960$ km.

Average speed. Total time $= 1 + 4 = 5$ h, so average $= 960 \div 5 = 192$ km/h.

> **Answer:** Whole journey $=$ **960 km**; average speed $=$ **192 km/h**.

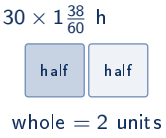
Practice. Use the bar model and the working space.

1 A cyclist rides for **4 h 44 min** at **15 km/h** and reaches the *halfway* point of a trail. How long is the whole trail?



Working space

2 A runner covers **1 h 38 min** at **30 km/h**, which is exactly half of a race. Find the full length of the race.



Working space

3 A car travels for **3 h 46 min** at **30 km/h** to reach the midpoint of a highway. How long is the whole highway?

$30 \times 3\frac{46}{60}$ h



whole = 2 units

Working space

4 Ben drives **2 h 20 min** at **36 km/h** and that is half of his whole journey. Find the whole distance.

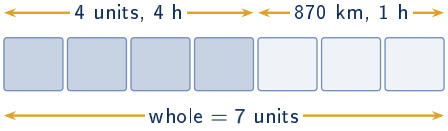
$36 \times 2\frac{20}{60}$ h



whole = 2 units

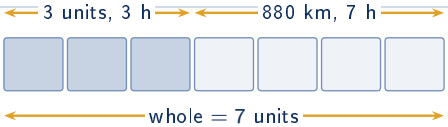
Working space

5 Sophia completes $\frac{4}{7}$ **of a journey in 4 h**. The remaining **870 km** takes just **1 h** on a fast road. Find the whole journey and her average speed.



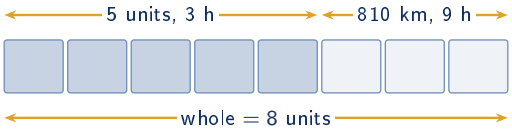
Working space

6 Lucas cycles $\frac{3}{7}$ **of a route in 3 h**, then the last **880 km** take **7 h**. Find the whole route and his average speed.



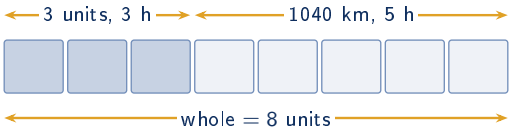
Working space

7 Ethan drives $\frac{5}{8}$ **of a trip in 3 h**. The remaining **810 km** take **9 h**. Find the whole trip and his average speed.



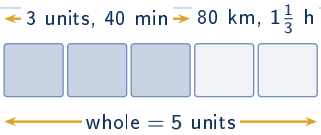
Working space

8 Emma covers $\frac{3}{8}$ **of a journey in 3 h**, then the last **1040 km** in **5 h**. Find the whole journey and her average speed.



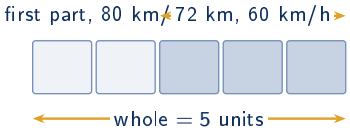
Working space

9 Sophia walks $\frac{3}{5}$ **of a path in 40 min**, then the last **80 km** take $1\frac{1}{3}$ **h**. Find the whole path and her average speed.



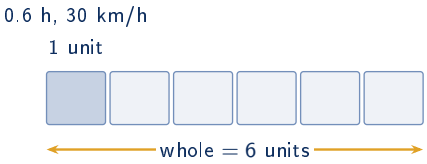
Working space

10 Ben drives from town A to town B. The *last* $\frac{3}{5}$ of the whole trip is **72 km**, driven at **60 km/h**. He drives the first part at **80 km/h**. Find the length of the first part and the total time.



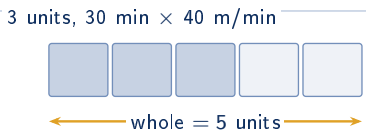
Working space

11 A train covers $\frac{1}{6}$ **of a line in 0.6 h** at **30 km/h**. How long is the whole line?



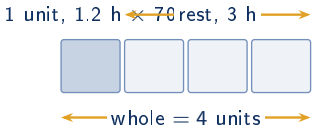
Working space

12 Emma walks for **30 min** at **40 m/min**, which is $\frac{3}{5}$ of her way to school. How far is school (in metres)?



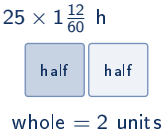
Working space

13 Bus A drives for **1.2 h at 70 km/h**, covering $\frac{1}{4}$ of a route. Bus B then drives the rest in **3 h**. Find the whole route and the average speed for the trip.



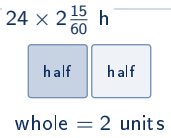
Working space

14 A scooter rides **1 h 12 min** at **25 km/h** to reach the halfway point of a lane. How long is the whole lane?



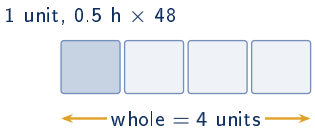
Working space

15 A hiker walks **2 h 15 min** at **24 km/h** (on wheels!) to reach the midpoint of a trail. Find the whole length.



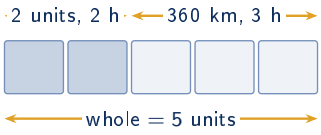
Working space

16 A car covers $\frac{1}{4}$ **of a journey in 0.5 h** at **48 km/h**. How long is the whole journey?



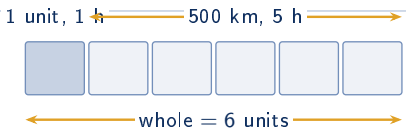
Working space

17 Lucas drives $\frac{2}{5}$ **of a journey in 2 h**, then the last **360 km** in **3 h**. Find the whole journey and his average speed.



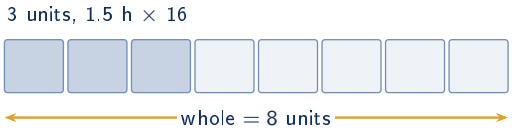
Working space

18 A coach covers $\frac{1}{6}$ **of a tour in 1 h**, then the remaining **500 km** in **5 h**. Find the whole tour and the average speed.



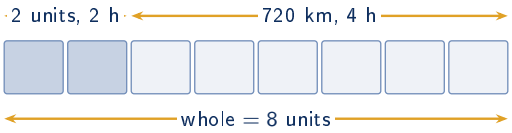
Working space

19 A train covers $\frac{3}{8}$ **of a line in 1.5 h** at **16 km/h**. How long is the whole line?



Working space

20 Oliver drives $\frac{1}{4}$ **of a journey in 2 h**, then the remaining **720 km** in **4 h**. Find the whole journey and his average speed.



Working space

Chapter 9

Journeys with a change of speed

Real trips rarely keep one speed. A driver slows for traffic, speeds up on a clear road, or returns home more slowly than they set out. The trick is always the same: *break the journey into legs*, handle each leg with the speed triangle, then combine. Keep distance, speed and time in matching units, and remember that a *round trip* covers the same distance each way.

Key idea — one leg at a time

Use the three forms of the speed relationship, whichever fits the leg you are on:

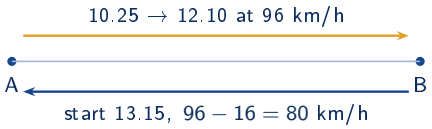
$$\text{distance} = \text{speed} \times \text{time}, \quad \text{time} = \frac{\text{distance}}{\text{speed}}, \quad \text{speed} = \frac{\text{distance}}{\text{time}}.$$

1. **Split** the trip into legs (out / back, slow / fast, first part / rest).
2. **Fill the gaps** leg by leg — a small table keeps distance, speed and time tidy.
3. **Combine.** For a there-and-back trip the two distances are equal. For an average, use $\text{average} = \dfrac{\text{total distance}}{\text{total time}}$.

Write clock times with a **dot**: 13.00, not "13 00". A "20% slower" speed means multiply by 0.8; "20% faster" means multiply by 1.2.

Example 1 — there and back

Ben drives from home (A) to a lake (B), leaving at **10.25** and arriving at **12.10** at **96 km/h**. After a rest he starts back at **13.15**, but traffic makes him **16 km/h** slower. At what time does he reach home?



Solution

Length of the trip (the "out" leg). From 10.25 to 12.10 is 1 h 45 min = 1.75 h, so $AB = 96 \times 1.75 = 168$ km.

Return speed. $96 - 16 = 80$ km/h.

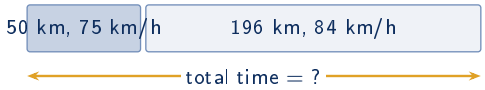
Return time. $168 \div 80 = 2.1$ h = 2 h 6 min.

Arrival time. 13.15 + 2 h 6 min = 15.21.

> **Answer:** He reaches home at **15.21**.

Example 2 — two legs, two speeds

Lucas drives the first **50 km at 75 km/h**, then another **196 km at 84 km/h**. How long does the whole journey take?



Solution

First leg. $50 \div 75 = \frac{2}{3}$ h $= 40$ min.

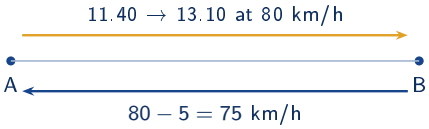
Second leg. $196 \div 84 = \frac{7}{3}$ h $= 2$ h 20 min.

Add the legs. 40 min $+ 2$ h 20 min $= 3$ h.

> **Answer:** The whole journey takes **3 h**.

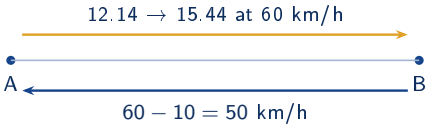
Practice. Sketch each trip, then use the working space.

1 Lucas leaves at **11.40** and arrives at **13.10** driving at **80 km/h**. He returns along the same road **5 km/h** slower. How long does the return journey take?



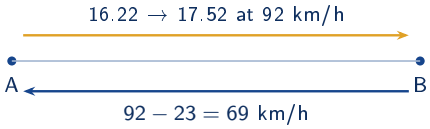
Working space

2 Emma leaves at **12.14** and arrives at **15.44** at **60 km/h**. Her return is **10 km/h** slower. How long does the return take?



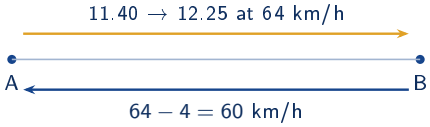
Working space

3 Oliver leaves at **16.22** and arrives at **17.52** at **92 km/h**. He comes back **23 km/h** slower. How long is the return journey?



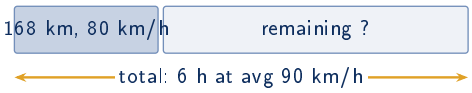
Working space

4 Sophia leaves at **11.40** and arrives at **12.25** at **64 km/h**. She returns **4 km/h** slower. How long does the return take?



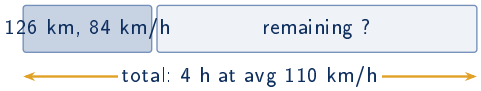
Working space

5 Emma plans a **6 h** drive at an average of **90 km/h**. She drives the first **168 km at 80 km/h**. At what speed must she cover the rest to keep her planned average? (1 d.p.)



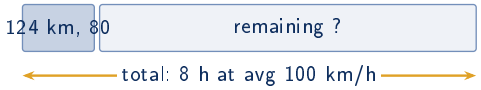
Working space

6 Ben drives for **4 h** at an average of **110 km/h**. The first **126 km** are at **84 km/h**. Find the speed needed for the rest. (1 d.p.)



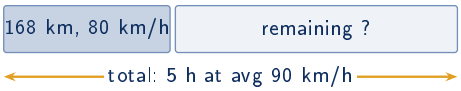
Working space

7 Ethan drives for **8 h** at an average of **100 km/h**. The first **124 km** are at **80 km/h**. Find the speed needed for the rest. (1 d.p.)



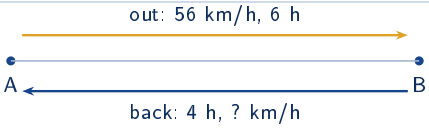
Working space

8 Sophia drives for **5 h** at an average of **90 km/h**. The first **168 km** are at **80 km/h**. Find the speed needed for the rest. (1 d.p.)



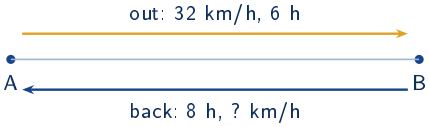
Working space

9 Oliver drives to a city in **6 h** at **56 km/h**, then returns along the same road in **4 h**. By how much do his two average speeds differ?



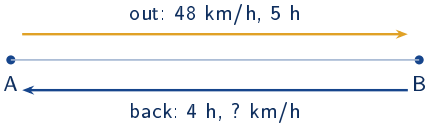
Working space

10 Ben drives out in **6 h** at **32 km/h** and comes back in **8 h**. By how much do his average speeds differ?



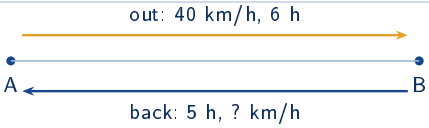
Working space

11 Emma drives out in **5 h** at **48 km/h** and returns in **4 h**. Find the difference in her average speeds.



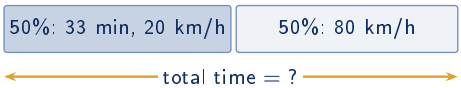
Working space

12 Lucas drives out in **6 h** at **40 km/h** and returns in **5 h**. Find the difference in his average speeds.



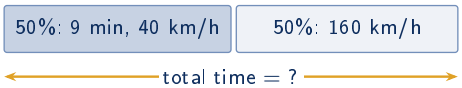
Working space

13 Sophia covers the **first 50%** of a trip in **33 min at 20 km/h**, then drives the rest at **80 km/h**. Find the total time. (1 d.p., in minutes)



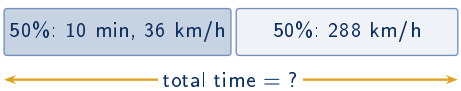
Working space

14 Ethan covers the **first 50%** of a trip in **9 min at 40 km/h**, then the rest at **160 km/h**. Find the total time. (1 d.p., in minutes)



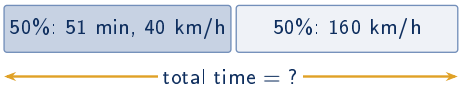
Working space

15 Emma covers the **first 50%** of a trip in **10 min at 36 km/h**, then the rest at **288 km/h** (a fast train!). Find the total time. (1 d.p., in minutes)



Working space

16 Oliver covers the **first 50%** of a trip in **51 min at 40 km/h**, then the rest at **160 km/h**. Find the total time. (1 d.p., in minutes)



Working space

17 Ethan drives the first $\frac{5}{9}$ of a journey at **80 km/h**. For the last $\frac{4}{9}$ he slows by **20%**, which takes him **2.5 h**. Find the total time for the journey.



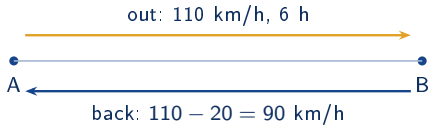
Working space

18 Emma drives the first $\frac{5}{9}$ of a journey at **80 km/h**. For the last $\frac{4}{9}$ she speeds up by **20%**, which takes her **2.5 h**. Find the total time for the journey.



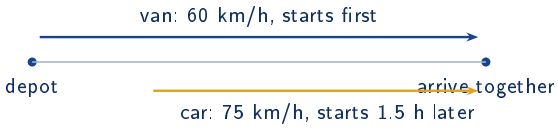
Working space

19 A coach drives for **6 h at 110 km/h** to a resort, then returns along the same road **20 km/h** slower. How long does the return journey take?



Working space

20 A van leaves a depot at **60 km/h**. A car sets off along the same road **1.5 h later** at **75 km/h**, and the two arrive at the destination at the same moment. For how long does the *car* travel?



Working space

End of supplement

ThinkAlgo • Speed • Chapters 8 & 9

Student edition